

NATURAL LANGUAGE PROCESSING (NLP) – INTRODUCTION

- **Subject:** Artificial Intelligence / NLP
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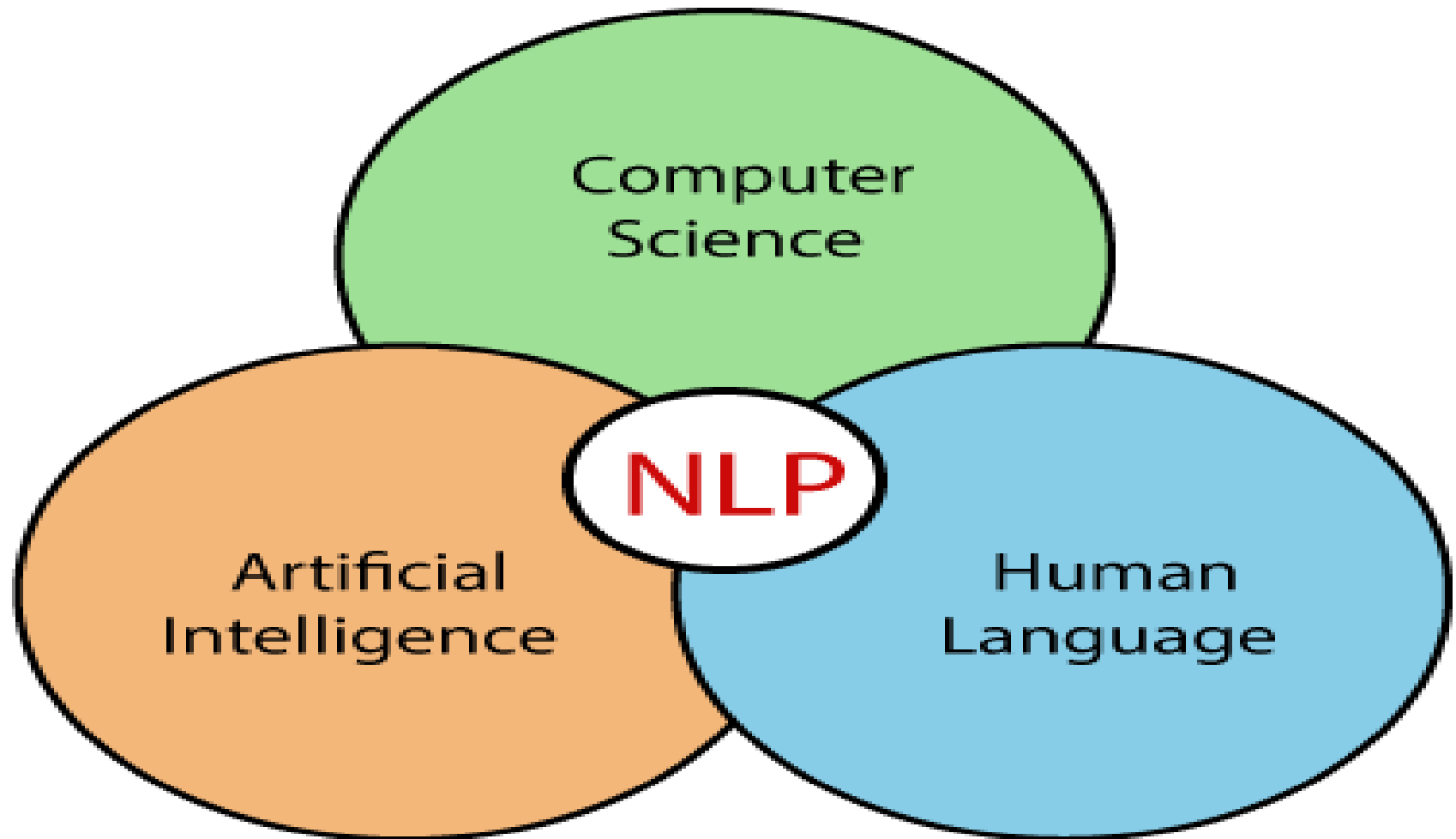
- N-grams, Evaluation, Smoothing, Interpolation, Backoff
- Word Classes, POS Tagging
- Rule-based, Stochastic, Transformation-based tagging
- Hidden Markov Models, Maximum Entropy Models

WHAT IS NLP?

Definition:

- Natural Language Processing (NLP) is a branch of Artificial Intelligence that enables computers to understand, interpret, and generate human languages such as English, Hindi, Telugu, etc.
- **In Simple Words:**
- NLP helps machines talk like humans and understand what humans speak or write.
- **Example:**
- Google Assistant replying to questions
- Spell checker in MS Word
- ChatGPT answering questions

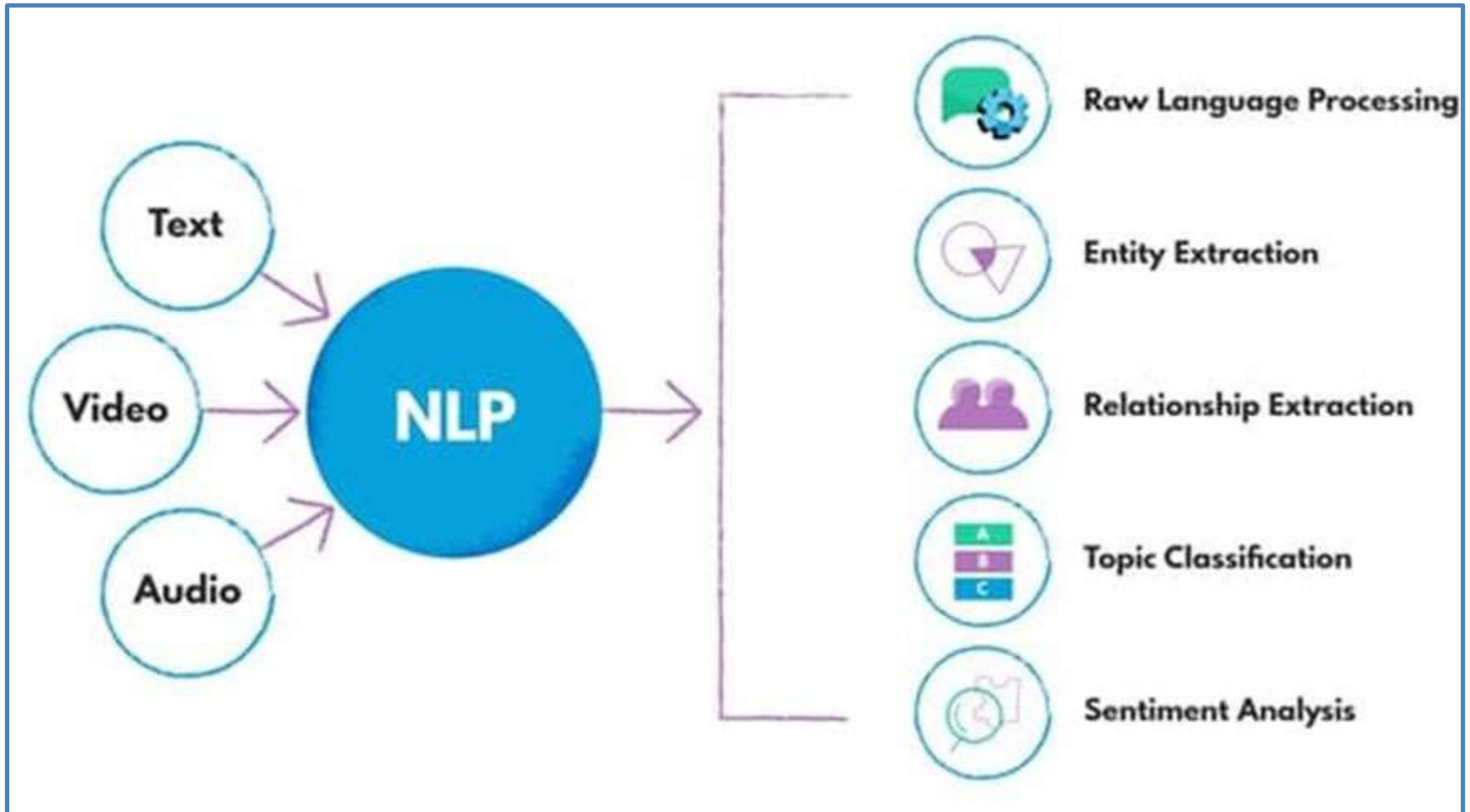
Overview of NLP



Why NLP is Needed?

- Computers understand only **numbers and programs**, but humans speak using **natural language**.
- NLP works as a **bridge** between humans and machines.
- **Example:**
- When you type:
- "Best mobile under 20,000"
- Google understands:
- You want a phone
- You are giving a price range
- You want recommendations

Need of NLP



COMMON APPLICATIONS

- Voice Assistants (Siri, Alexa)
- Machine Translation (Google Translate)
- Spam detection in emails
- Chabot's and virtual agents
- Sentiment analysis (opinions on social media)
- Text summarization and question answering.

Challenges in NLP (Difficulties in Language Processing)

- Understanding human language is difficult because:
- A word can have more than one meaning
- Sentence meaning depends on context
- Grammar varies
- People mix languages
- Language keeps changing

AMBIGUITY IN NLP

- What is Ambiguity?
- Ambiguity means a word, sentence, or phrase has **more than one meaning** and the computer becomes confused about the correct meaning.

1. LEXICAL AMBIGUITY

One word has many meanings.

Examples:

1.The word **bat** can mean:

- Cricket bat
- Animal (bat)

2.The word **light** can mean:

- Not heavy
- Bright

2. SYNTACTIC AMBIGUITY

The sentence structure is unclear.

- **Example:**
- "I saw the man with a telescope."
- Meaning 1:
You used the telescope to see the man.
- Meaning 2:
The man had the telescope.

3. SEMANTIC AMBIGUITY

The meaning of the sentence is not clear.

- **Example:**
- "I saw her duck."
- Duck could mean:
- A bird
- Bending action
- The computer does not know the correct meaning without context.

4. PRAGMATIC AMBIGUITY

Meaning depends on the situation.

Example:

- "Can you pass the salt?"
- It sounds like a question, but actually it is a request.
- "He is hot."
Could mean:
 - Weather is hot
 - Person looks attractive

5. REFERENTIAL AMBIGUITY

It is not clear who or what is being talked about.

- **Example:**

"Ravi met Arjun when he was leaving."

- Who was leaving?
- Ravi?
- Arjun?

Another example:

"The students spoke to teachers because they were confused."

Who was confused?

- Students?
- Teachers?

B) VARIABILITY OF LANGUAGE

same meaning can be expressed in many ways:

- Example:
- "I'm fine"
"I'm okay"
"I'm good"
- Spelling changes:
colour / color
- Code mixing:
"Let's go chalo"

C) Context Dependence

Meaning changes with situation

- Examples:
- “The trophy didn't fit because it was big.”
- “Close the window; it's cold.”

d) Common Sense Reasoning

- Computers lack real-world logic.

✓ Example:

"The trophy didn't fit because it was too big."
Humans know "it" = trophy, not suitcase.

e) Sentiment Understanding

Hard to detect:

- Emotions
- Sarcasm
- Tone

Example:

"Wow, that's just great 😊"

Human understands it is sarcasm, but a machine may think it's positive.

Linguistic Diversity

- Different accents, slang, dialects and newly created words make understanding harder.
- **Example:**
Words like:
 - "Chill" = relax
 - "Lit" = excitingDifferent regions use different meanings → confusion for machines.

f) Data Limitations

- Example:
A voice assistant trained only on American English may not understand Indian accents correctly.

g) Multilingual Processing

- Example:
English: "I eat apples."
Hindi: "मैं सेब खाता हूँ"
Word order and grammar are different → Hard to handle for systems.

Challenges in NLP

Ambiguity



Data Scarcity



Contextual Understanding



Bias



ORIGINS OF NLP (EARLY HISTORY)

- Natural Language Processing began as an idea to make computers understand human language.
- **Early Foundations:**
 - 1950 – Alan Turing proposed the **Turing Test** to check machine intelligence.
 - 1954 – First machine translation experiment (English ↔ Russian).
 - First focus was language translation.
 - Computers worked using fixed grammar rules and small dictionaries.
- **Problem:**
 - Rule-based systems worked only for limited sentences and could not understand meaning well.

ORIGINS OF NLP (MODERN DEVELOPMENT)

- As computers improved, NLP techniques changed.

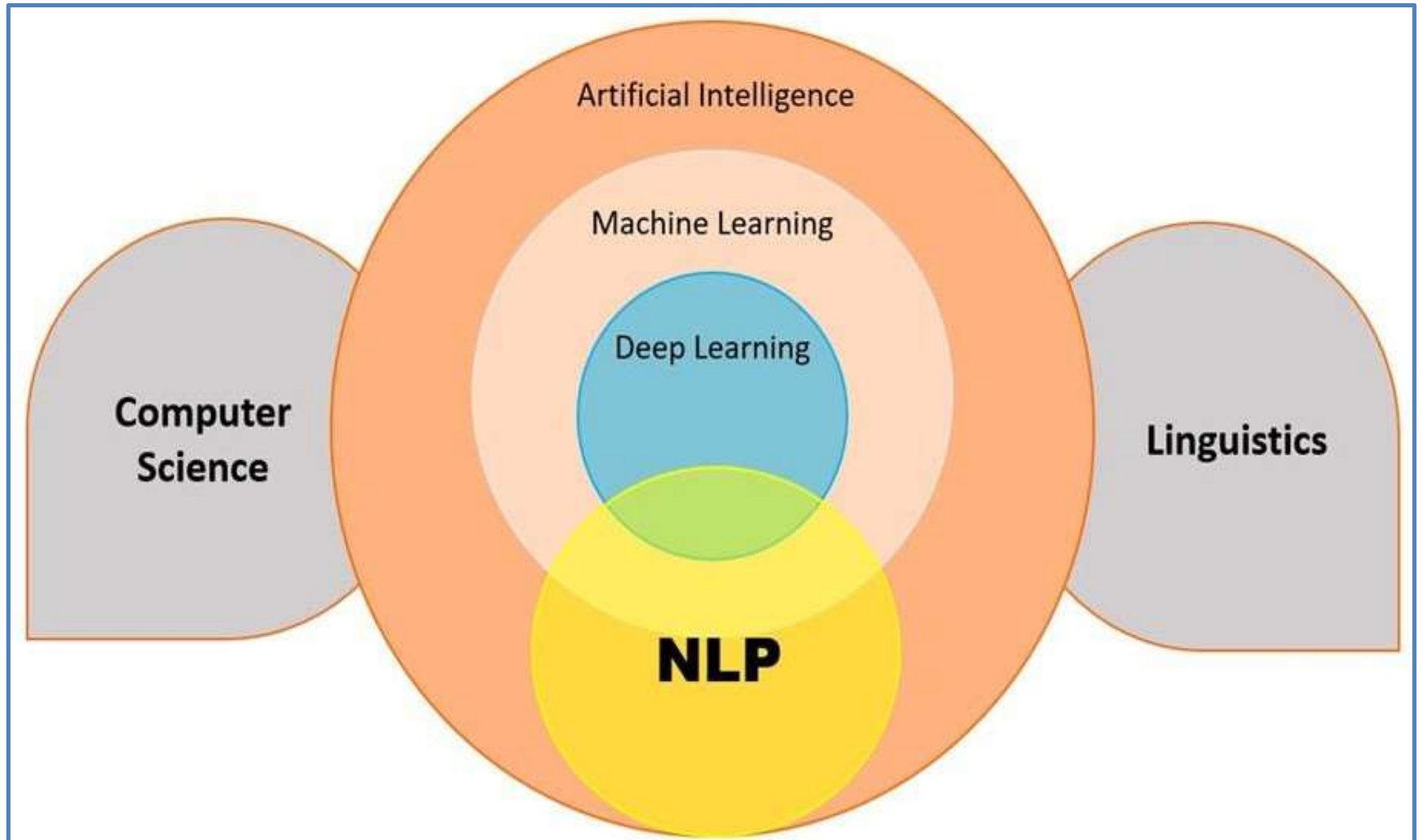
Later Development:

- 1970s–80s – Knowledge-based systems (meaning + logic).
- 1990s – Statistical NLP (learning from data).
- 2000s – Machine Learning introduced.
- 2010–Present – Deep learning & AI models (BERT, GPT).

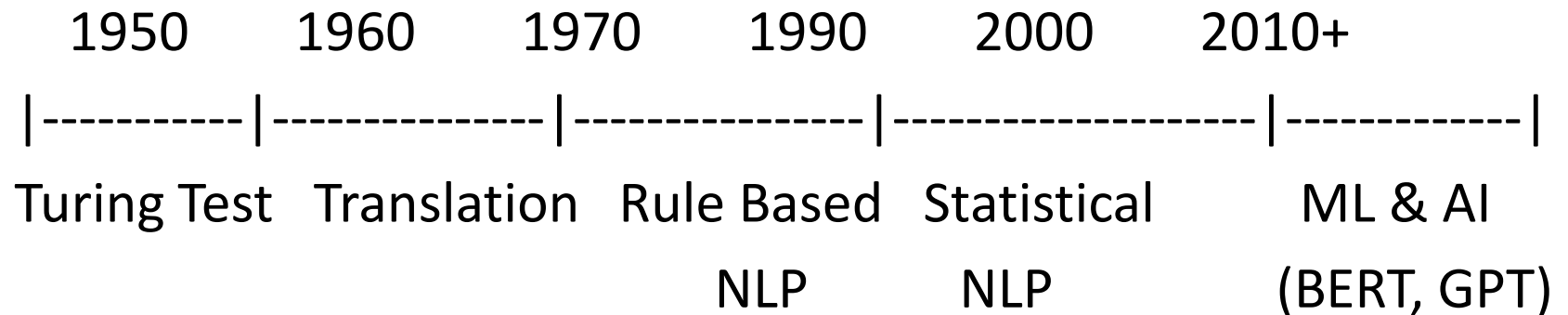
Result:

- NLP became:
 - Fast
 - Accurate
 - Intelligent
 - Human-like
- Examples: ChatGPT, Google Assistant, Alexa.

Origins of NLP



TimeLine Diagram



LANGUAGE MODELLING IN NLP

- Language Modelling is the process of teaching computers how human language is structured and how words usually appear together.
- In simple terms:
A Language Model helps a computer decide **which sentence is correct** and **what word comes next**.

TYPES OF LANGUAGE MODELS

There are three main ways to model language:

- Grammatical Language Models
- Statistical Language Models
- Regular Expressions (Regex-based pattern model)

GRAMMATICAL LANGUAGE MODEL

- Grammatical Language Models use **grammar rules** to understand language.
- They check whether a sentence:
 - follows grammar
 - is structurally correct
- These models use rules like:
Subject → Verb → Object

CONTEXT-FREE GRAMMARS (CFG)

✓ Definition:

- Grammar rules to form valid sentences.

✓ Common Non-terminals:

- $S \rightarrow$ Sentence
- $NP \rightarrow$ Noun Phrase
- $VP \rightarrow$ Verb Phrase

☆ Sample CFG Rules:

GIVEN GRAMMAR RULES

- $S \rightarrow NP VP$
 $NP \rightarrow Det N$
 $VP \rightarrow V NP \mid V$
 $Det \rightarrow the \mid a$

 $N \rightarrow boy \mid girl \mid apple$
 $V \rightarrow eats \mid likes$

WHAT EACH SYMBOL MEANS

- S = Sentence
NP = Noun Phrase
VP = Verb Phrase
Det = Determiner (article)
N = Noun
V = Verb

RULE-BY-RULE EXPLANATION

1. $S \rightarrow NP VP$

A sentence is made of:
Noun Phrase + Verb Phrase

2. $NP \rightarrow Det N$

A noun phrase is:
Article + Noun

- Examples:
the boy
a girl
the apple

3. VP $\rightarrow$ V NP | V

A verb phrase can be: Verb + Noun Phrase
or
Only Verb

4. Det $\rightarrow$ the | a

Determiners are: the, a

5. N $\rightarrow$ boy | girl | apple

Possible nouns are: boy, girl, apple

6. V $\rightarrow$ eats | likes

Possible verbs are: eats, likes

HOW A SENTENCE IS FORMED

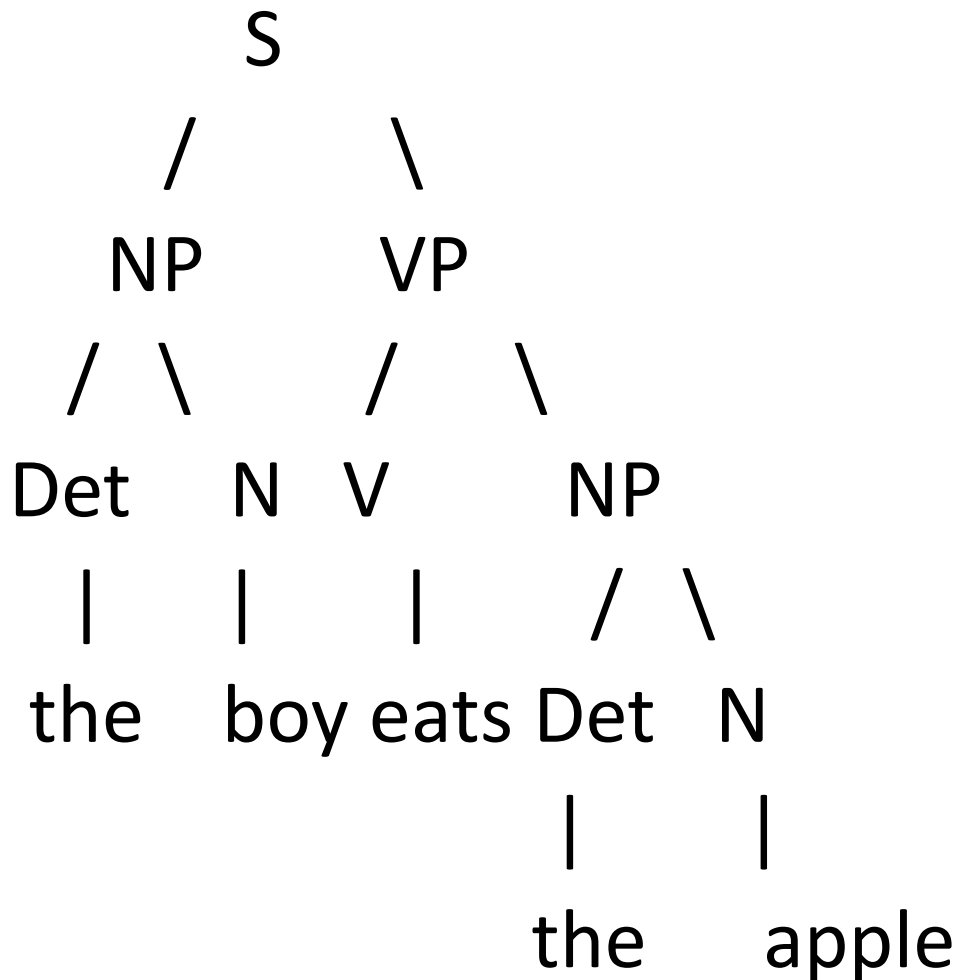
Let us create a sentence using the grammar.

Example 1:

- Sentence: The boy eats the apple
- Use rules:
- $S \rightarrow NP VP$
 $NP \rightarrow \text{the boy}$
 $VP \rightarrow \text{eats the apple}$
- So full sentence:
 The boy eats the apple
- ✓ Grammatically correct

- Example 2:
- Sentence: A girl likes the boy
- $S \rightarrow NP VP$
NP $\rightarrow$ a girl
VP $\rightarrow$ likes the boy
- ✓ Correct sentence
- Example 3 (Incorrect):
- "Eats boy apple"
- ✗ Incorrect because grammar rule is broken.

Parse tree



STATISTICAL LANGUAGE MODEL (SLM)

- A Statistical Language Model predicts words using probability and data instead of grammar rules.
- In simple words:
It learns from examples and decides which sentence is most likely to be correct.

HOW STATISTICAL LM WORKS

- Statistical LM learns from **large text data (corpus)**.

It counts:

- How often words appear
- Which words come together
- Which word follows another
- Based on this, it predicts the next word

SIMPLE EXAMPLE

- Sentence:
"I am very ____"
- Prediction based on data:
happy
tired
hungry
- The model chooses the word with highest probability.

BASIC FORMULA

- Statistical LM calculates:
- $P(\text{sentence}) = \text{Probability of word sequence}$
- Example:
 $P(\text{"I love coding"}) = P(\text{coding} \mid \text{I love})$

TYPES OF STATISTICAL LM

Statistical LM uses:

- Unigram model
- Bigram model
- Trigram model

They are called N-gram Models.

Formulas:

- 1) unigram ($n=1$) = $P(w)$ contains one word
- 2) bigram ($n=2$) = $P(w_2/w_1)$ contains two words
- 3) trigram ($n=3$) = $P(w_3/w_1, w_2)$ contains 3 words.

UNIGRAM MODEL

- Assumption:
Each word is independent.
- Example:
"I like NLP"
- Probability:
 $P(I) \times P(\text{like}) \times P(\text{NLP})$
- Problem:
Word order is ignored.

BIGRAM MODEL

- Here each word depends on the previous one.
- Example:
"Good morning"
- Probability:
 $P(\text{morning} \mid \text{good})$
- Better accuracy than unigram.

TRIGRAM MODEL

- Each word depends on two previous words.
- Example:
"I am learning"
- Probability:
 $P(\text{learning} \mid \text{I am})$
- Most accurate but needs more data.

N-GRAM MODELS WITH CLEAR CORPUS EXAMPLES

- Corpus Used:
“The cat sat on the mat”
- Tokenization:
splitting text into words or sentences
Ex: “The cat sat on the mat” = [the, cat, sat, on, the, mat]

Unigram model

✓ Word Counts:

Word	Count	Probability
The	2	$2/6 = 0.33$
cat	1	$1/6 = 0.16$
sat	1	$1/6 = 0.16$
on	1	$1/6 = 0.16$
mat	1	$1/6 = 0.16$

☆ Example:

- Probability of “the cat sat”
- $P(\text{the}) \times P(\text{cat}) \times P(\text{sat})$

$$= 0.33 \times 0.16 \times 0.16$$

$$= 0.0084$$

Bigram model

✓ Bigrams:

the cat
cat sat
sat on
on the
the mat

✓ Probabilities:

Bigram	Count	P
the cat	1	$1/2 = 0.5$
cat sat	1	$1/1 = 1$
sat on	1	$1/1 = 1$
on the	1	$1/1 = 1$
the mat	1	$1/2 = 0.5$

☆ Example:

Probability of “the cat sat” $0.5 \times 1 = 0.5$

REGULAR EXPRESSIONS (REGEX) IN NLP

- Regular Expressions (Regex) are patterns used to search, match, and extract text.
- In simple words:

Regex helps NLP systems find specific patterns in text.

WHY REGEX IS IMPORTANT IN NLP

Regex is used to:

- Find particular words
- Check format of input
- Clean text data
- Extract useful information
- It works faster than training a model for pattern tasks.

BASIC REGEX SYMBOLS

- [0–9] → matches digits
- [a–z] → matches lowercase letters
- [A–Z] → matches uppercase letters
- . → any single character
- * → repeat zero or more times
- + → repeat one or more times
- ^ → start of string
- \$ → end of string

SIMPLE REAL-WORLD USES

- Regex is used in:
- Email validation
- Phone number checking
- Password checking
- Search filters
- Log file analysis
- Web scraping
- Example:
Checking if email format is correct
name@gmail.com ✓
name.gmail ✗

Use in NLP	Meaning	Regex Example	Real-life Example
Tokenization	Splitting text into words	\w+	"I love NLP" → I, love, NLP
Removing punctuation	Cleans text	[^\w\s]	Removes .,!? from sentences
Finding numbers	Detect digits	\d+	Extract "2024" from "Year is 2024"
Email extraction	Extract email IDs	\S+@\S+\.\S+	Gets abc@gmail.com from text
URL detection	Find websites	https?:\/\/\S+	Extract links
Phone numbers	Detect phone formats	\d{10}	Finds 10-digit mobile numbers
Whitespace cleanup	Remove extra spaces	\s+	Converts multiple spaces to one
Find capital words	Find capital words	Find capital words	Find capital words
Text validation	Check format	^[A-Za-z]+\$	Confirms only alphabets
Date extraction	Find dates	\d{2}\/\d{2}\/\d{4}	Finds "12/05/2025"

EXTRACTING INFORMATION USING REGEX

- From sentence:
"Contact: ramya@gmail.com"
Regex extracts:
Email ID only.
- From text:
"Call me at 9876543210"
Regex extracts:
Phone number.

FINITE STATE AUTOMATA (FSA)

- A Finite State Automaton (FSA) is a machine that reads input symbols one by one and decides whether a string is correct or not.
- In simple words:
FSA checks whether a given word or pattern follows the rules.

WHY FSA IS USED IN NLP

FSA is used to:

- Recognize words
- Check valid patterns
- Process text step by step
- Detect spelling formats
- Used in:
Search filters, text processing, simple language recognition.

PARTS OF FSA

An FSA contains:

- States (different positions in processing)
- Input symbols (letters or digits)
- Start state (where machine begins)
- Final state (accepted result)
- Transition rules (movement between states)

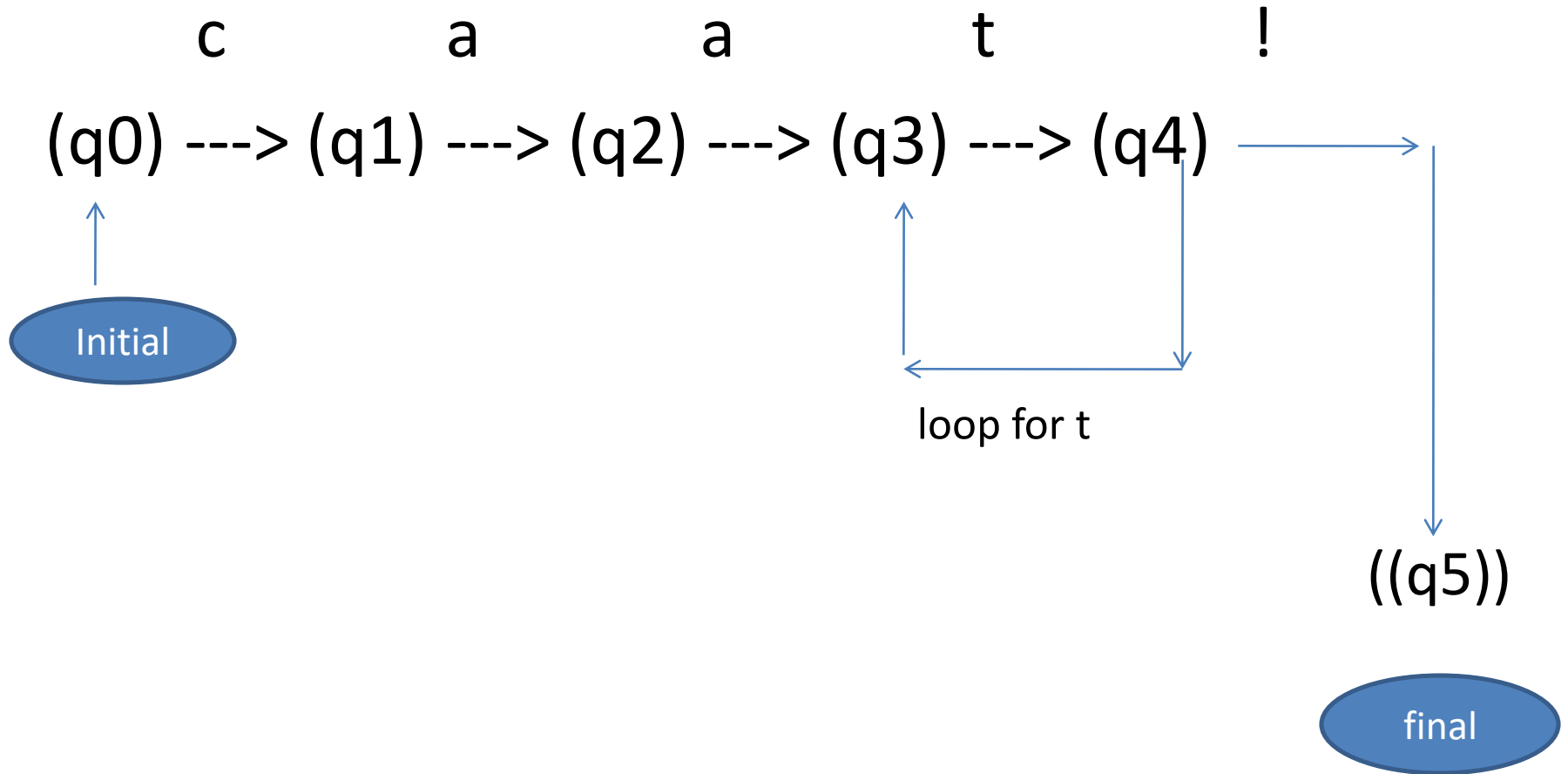
HOW FSA WORKS

- 1.FSA starts at the start state.
- 2.It reads one character at a time.
- 3.For each character, it moves to another state.
- 4.If it finally reaches an accept state, the input is correct.
Otherwise, it is rejected.

Example: caat+!

Symbol	Meaning
c	Must appear once
a	Must appear once
a	Must appear once
t+	One or more t's
!	Must end with !
+	Means “one or more occurrences”

State transition diagram



State transition table

State	c	a	t	!
q0	q1	ϕ	ϕ	ϕ
q1	ϕ	q2	ϕ	ϕ
q2	ϕ	q3	ϕ	ϕ
q3	ϕ	ϕ	q4	ϕ
q4	ϕ	ϕ	q4	q5
q5	ϕ	ϕ	ϕ	ϕ

ENGLISH MORPHOLOGY

- Morphology is the study of how English words are formed and changed.
- In simple words:

Morphology explains how words are built using parts like roots, prefixes, and suffixes.

WHAT IS A MORPHEME?

- A morpheme is the smallest meaningful part of a word.

Examples:

- book → 1 morpheme
- books → book + s (2 morphemes)
- unhappy → un + happy

TYPES OF MORPHEMES

Free morpheme

- Can stand alone as a word

Examples:

boy, run, happy, book

Bound morpheme

- Cannot stand alone

Examples:

un-, -ness, -ed, -ing

ROOT, PREFIX, SUFFIX

- Root word: main part
Prefix: comes before root
Suffix: comes after root
- Examples:
un + happy = unhappy
play + ing = playing
kind + ness = kindness

TYPES OF MORPHOLOGY IN ENGLISH

There are 2 main types:

- Inflectional morphology
- Derivational morphology

INFLECTIONAL MORPHOLOGY

- Word form changes but meaning remains the same.
- Examples:
 - run → running
 - boy → boys
 - tall → taller
 - good → better
 - go → went
- Purpose:
 - Shows tense, number, degree, etc.
- Used in:
 - Grammar correction tools.

DERIVATIONAL MORPHOLOGY

A new word is formed and meaning changes.

- Examples:
happy → happiness
teach → teacher
nation → national
quick → quickly
act → reaction
- Purpose:
Creates new vocabulary.

MORPHOLOGY IN NLP

In NLP, morphology helps computers:

- Understand word forms
- Identify root words
- Convert tenses
- Handle plurals
- Improve translation
- Example:
studies → study
played → play

LIMITATIONS

- Irregular words
- Exceptions in English
- Difficult rules
- Slang confusion

TRANSDUCERS FOR LEXICONS

- A Transducer is a machine that converts one form of a word into another form.
- A Lexicon is a dictionary that stores words.
- So, Transducers for Lexicons means using special machines to change word forms automatically using dictionary knowledge.

WHAT IS A FINITE STATE TRANSDUCER (FST)?

A Finite State Transducer (FST) is a type of machine that:

- Reads a word
- Converts it into another word
- Produces output

In simple words:

An FST takes input word and gives output word.

WHY TRANSDUCERS ARE USED FOR LEXICONS

Transducers help with:

- Handling plural forms
- Handling verb tenses
- Word analysis
- Word generation
- Dictionary mapping

HOW FST WORKS

An FST:

- Takes a word as input
- Looks at the dictionary (lexicon)
- Applies language rules
- Produces correct form of the word

Example:

Input: play

Rule: add -ed

Output: played

MORPHOLOGY WITH FST

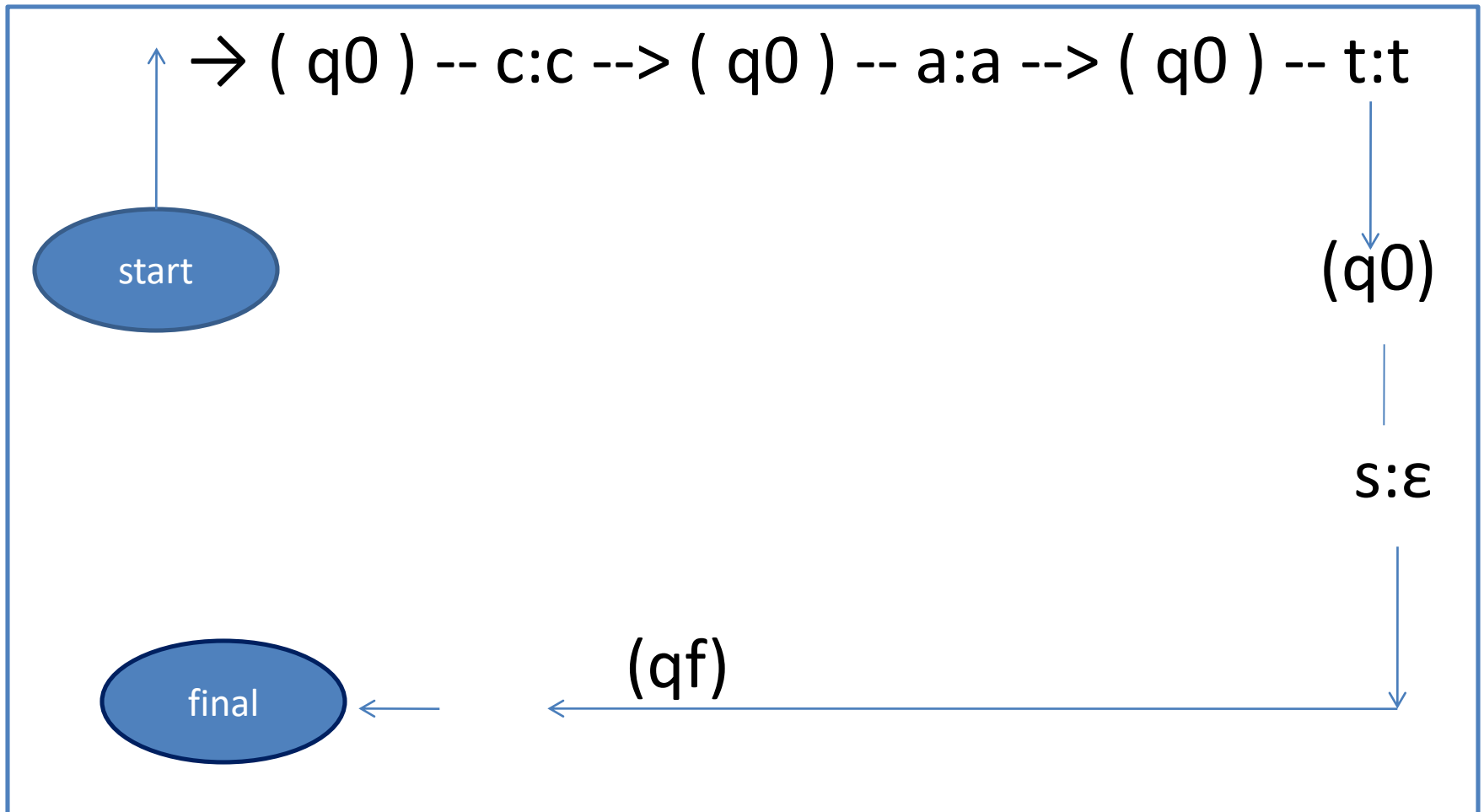
Transducers help in:

- Singular → plural
- Present → past
- Adjective → adverb
- Example:
happy → happiness
quick → quickly

Task:

Convert plural noun to singular.

cats $\rightarrow$ cat



FSA vs FST

- FSA:
Checks the word
- FST:
Changes the word
- Example:
FSA checks if "playing" is valid.
FST converts "play" into "playing".

TOKENIZATION IN NLP

- Tokenization is the process of breaking text into small units called *tokens*.
- Tokens can be:
 - Words
 - Sentences
 - Characters
 - Sub-words
- In simple words:
Tokenization helps a computer read text word by word instead of as one long sentence.

WHY TOKENIZATION IS IMPORTANT

- Computers cannot understand raw text. They must first convert text into pieces.
- Tokenization helps in:
- Understanding word meaning
- Searching text
- Text analysis
- Machine learning tasks
- Sentence understanding

TYPES OF TOKENIZATION

There are mainly 4 types:

- Word Tokenization
- Sentence Tokenization
- Character Tokenization
- Sub-word Tokenization

WORD TOKENIZATION

- Splits text into words.
- Example:
"I am learning NLP"
- Tokens:
I | am | learning | NLP
- Used in:
Search engines, chatbots, spell checkers.

SENTENCE TOKENIZATION

- Splits text into sentences.
- Example:
"I love NLP. It is interesting."
- Sentences:
I love NLP
It is interesting
- Used in:
Document summarization, text analysis.

CHARACTER TOKENIZATION

- Splits text into characters.
- Example:
"cat"
- Tokens:
c | a | t
- Used in:
Speech recognition and handwriting recognition.

SUB-WORD TOKENIZATION

- Breaks words into smaller meaningful parts.
- Example:
unhappy
un | happy
- Used in:
Neural machine translation and deep learning models.

TOKENIZING PUNCTUATION

- Sentence:
"Hello, how are you?"
- Tokens:
Hello | , | how | are | you | ?
- Why punctuation matters:
It helps NLP detect questions, emotions, and tone.

CHALLENGES IN TOKENIZATION

- Tokenizing is not always simple:
- Short forms:
don't = do + not
- Special symbols:
- @ \$ %
- Different languages:
Some languages do not use spaces
- Mixed languages:
"Let's go chalo"

DETECTING AND CORRECTING ERRORS IN NLP

- Error detection and correction is the process of identifying mistakes in text and suggesting the correct form.
- In simple words:
NLP helps find and fix **spelling, grammar and typing errors**.

WHY ERROR DETECTION IS IMPORTANT

- Errors reduce clarity.
- NLP correction:
- Improves communication
- Makes text professional
- Helps learners write better
- Improves search results
- Example:
Spell mistakes in Google still give correct output.

TYPES OF ERRORS

NLP mainly detects:

- Spelling errors
- Grammar errors
- Typing errors
- Word misuse
- Sentence structure errors

SPELLING ERRORS

- Wrong word length or letters.
- Examples:
microsft → microsoft
recieve → receive
langauge → language
- NLP suggests correct spelling using:
Dictionary and similarity logic.

GRAMMAR ERRORS

- Sentence structure wrong.
- Examples:
He go to school ✗
He goes to school ✓
- I eating food ✗
I am eating food ✓

TYPING ERRORS

- Errors by mistyping.
- Examples:
teh → the
wrod → word
- Apps fix automatically.

WORD MISUSE

- Wrong word selection.
- Examples:
Their / There / They're
Accept / Except
- NLP helps by understanding sentence meaning.

HOW NLP DETECTS ERRORS

NLP uses:

- Dictionary matching
- Regular expressions
- Minimum edit distance
- Language models
- Grammar rules

MINIMUM EDIT DISTANCE (MED) IN NLP

- Minimum Edit Distance (MED) is a method to find how *different* two words are.
- In simple words:
It tells how many changes are needed to convert one word into another.

WHY MED IS IMPORTANT

MED is used to:

- Find spelling mistakes
- Suggest correct words
- Compare similarity between words
- Improve search results
- Autocorrect typing

EDIT OPERATIONS

There are three operations:

- Insert – add a letter
- Delete – remove a letter
- Substitute – replace a letter

SIMPLE EXAMPLES

- Example 1:
cat $\rightarrow$ cut
Change: a $\rightarrow$ u
MED = 1
- Example 2:
book $\rightarrow$ back
Change: o $\rightarrow$ a, o $\rightarrow$ c
MED = 2
- Example 3:
kitten $\rightarrow$ sitting
Changes:
k $\rightarrow$ s
e $\rightarrow$ i
g added
MED = 3

HOW MED WORKS

The algorithm:

- Compares letters
- Finds the cheapest set of changes
- Chooses minimum number of edits
- Result:
Smaller number = words are more similar.

WEIGHTED EDIT DISTANCE

- Sometimes, some changes cost more.
- Example:
Changing "a $\rightarrow$ s" may cost less than "a $\rightarrow$ k"
- Used for:
Smart correction systems.

LIMITATIONS

- Fails in long words sometimes
- Slow for large text
- Does not check meaning
- Only works on letters, not sense

Thank
you!!!

